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6 ABSTRACT

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8 Introduction

9 GAP: No presynaptic model has taken into account temperature differences in release probability only calcium¹.

10 Model

11 Transitions between the presynaptic vesicle pools

12 Three pools of vesicles:

- 13 • Recycling pool
- 14 • Docked pool (i.e. readily releasable pool RRP)
- 15 • Fused pool

16 Calcium-dependent release probability (from docked to fused):

$$p_{rel} =$$

17 Calcium dynamics

$$\begin{aligned}\frac{dCa_{pre}}{dt} &= -\frac{Ca_{pre}}{\tau_{pre}} \\ \frac{dCa_{jump}}{dt} &= \frac{1 - Ca_{jump}}{\tau_{rec}} - \delta_{Ca} \cdot Ca_{jump} \cdot Ca_{pre} \\ Ca_{pre} &\leftarrow Ca_{pre} + Ca_{jump} \text{ upon spike arrival}\end{aligned}$$

Table 1. Model variables.

Symbol	Description	Unit
R	Number of vesicles in the recycling pool	-
D	Number of docked vesicles	-
F	Number of fused vesicles	-
Ca_{pre}	Intracellular presynaptic calcium concentration	mM
Ca_{jump}	Jump in Ca_{pre} upon spike arrival	mM

Table 2. Model parameters.

Symbol	Description	Unit	Reference values
D_0	Docked pool capacity	-	5-20 vesicles per bouton in cultured rat hipp. neurons ^{2,3}
R_0	Recycling pool capacity	-	15-40 vesicles per bouton in cultured rat hippocampal neurons (~3x larger than the docked pool capacity)
$\tau_{R \rightarrow D}$	Vesicle transition rate from recycling to docked	s	
$\tau_{D \rightarrow R}$	Vesicle transition rate from docked to recycling (mixing)	s	
$\tau_{F \rightarrow R}$	Vesicle transition rate from fused to recycling	s	
τ_{pre}	Ca_{pre} attenuation recovery rate	s	
τ_{rec}	Calcium jump recovery rate	s	
δ_{Ca}	Jump attenuation fraction	$s^{-1} mM^{-1}$	
Sigmoid parameters			

Table 3. Datasets.

Reference	Stim. conditions	Tissue	Age	Method	aCSF
Fernández-Alfonso T. et al. 2008 ⁴	300AP at 2, 5, 10, 20, 30 Hz 1800AP at 30 Hz (Baf, 0.5 μM) (for both 25C and 35C)	Rat cult. hipp. (autaptic)	P1-3	spHluorin (pH imaging)	119 NaCl, 2.5 KCl, 2 CaCl₂ , 2 MgCl ₂ , 30 glucose 25 HEPES (buf.pH 7.4) 10 μM CNQX and 50 μM AP5
delvendahl2016fast ⁵	temperature 25, 30 and 35	Mice cult. hipp. (autaptic)			

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